

Rising Waters

Flood Management System — AI/ML Project

Proposed Solution

Phase 3: Project Design Phase

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

Solution Overview

Rising Waters is a Machine Learning-powered flood prediction system that uses four classification algorithms (Decision Tree, Random Forest, KNN, XGBoost) trained on 5,000 meteorological records. The best-performing model — XGBoost at 96.55% accuracy — is saved as floods.save and deployed within a Flask web application.

Key Solution Components

- **Data Layer:** 5,000-row synthetic flood dataset with 6 meteorological features
- **ML Layer:** 4 algorithms trained, XGBoost selected (96.55% accuracy)
- **Application Layer:** Flask routes handling prediction, logging, history
- **Presentation Layer:** 6 premium dark-theme HTML pages (Jinja2)
- **Deployment Layer:** Docker + IBM Cloud for cloud hosting

Model Performance Summary